

RETAIL SALES & INVENTORY INTELLIGENCE SYSTEM

FINAL PROJECT REPORT

Excel | SQL | Python

Domain	Retail, Sales & Inventory Analytics
Data Period	January 2016 – December 2018
Total Records	5,000+ transactions across 9 tables
Stores	3 Stores — New York, California, Texas
Tools Used	Excel, SQL, Python (Pandas, Scikit-learn)

1. Executive Summary

This report presents the findings of a comprehensive data analysis project conducted for a retail company operating three stores across New York, California, and Texas. The company specialises in bicycles and related accessories, selling products across seven categories from nine global brands.

The analysis covered over 5,000+ transactions from January 2016 to December 2018 using a relational dataset of nine interconnected tables. The project was executed in four phases: data cleaning in Excel, database management and querying in SQL, exploratory and predictive analysis in Python, and visualisation-ready output for Power BI dashboards.

Key Headline Metrics

\$6,662,615 Total Net Revenue	1,445 Completed Orders	1,445 Total Customers	6,318 Units Sold
10.56% Avg Discount	13,511 Total Stock Units	458 Delayed Shipments	9 Active Staff

The analysis reveals that Baldwin Bikes in New York is the dominant revenue driver, Trek is the leading brand, Mountain Bikes lead in profitability, and significant opportunities exist in reducing delayed shipments, managing inventory gaps, and retaining high-value customers.

2. Project Overview

2.1 Problem Statement

The retail company required a comprehensive, end-to-end data analytics solution to improve decision-making across its operations. Key challenges included: lack of visibility into cross-store sales performance, no structured reporting on staff productivity, inability to track inventory health in real time, and limited insight into customer purchasing behaviour.

2.2 Business Use Cases

- Identify top-selling brands by region and store
- Evaluate staff performance based on total sales handled
- Track customer orders and their fulfilment status
- Identify the most profitable product categories
- Analyse stock levels across stores to optimise inventory
- Understand order trends across time (daily, weekly, monthly)
- Monitor delayed shipments versus required delivery dates
- Discover customer concentration and demographics
- Segment customers based on order volume and frequency (Optional)

2.3 Dataset Overview

The dataset consists of nine relational tables with structured CSV data spanning three years:

Table Name	Rows	Description
orders	1,615	Customer orders with status, dates, store, and staff
order_items	4,722	Line items per order with price, quantity, and discount
customers	1,445	Customer profiles with contact and location details
products	321	Product catalogue with brand, category, model year, and price
stores	3	Store profiles with location information
staffs	10	Staff records with store assignment and manager hierarchy
stocks	939	Stock quantity per product per store
brands	9	Brand reference table
categories	7	Product category reference table

3. Methodology & Approach

Phase 1 — Excel: Data Preprocessing

All nine raw CSV files were loaded and inspected in Excel. The following preprocessing steps were applied:

- Null handling: Missing phone numbers (1,267 records) filled with 'N/A'; null shipped_date left intact (170 records represent pending/rejected orders)
- Date formatting: order_date, required_date, and shipped_date standardised to YYYY-MM-DD format using pandas datetime parsing
- Derived columns: revenue = list_price x quantity x (1 - discount); order_status_label mapped from numeric codes (1=Pending, 2=Processing, 3=Rejected, 4=Completed); full_name concatenated for customers and staff; delay_days computed as shipped_date minus required_date
- Deduplication check: No duplicate order_id or customer_id found
- Foreign key validation: All join keys verified for referential integrity
- Output: Nine cleaned CSV files and one merged master CSV (4,722 rows x 37 columns)

Phase 2 — SQL: Database Management & Querying

Cleaned datasets were imported into a relational SQL database (MySQL/MariaDB compatible). The following was implemented:

- Full schema with PRIMARY KEY and FOREIGN KEY constraints across all nine tables
- Integrity checks confirming zero orphan records across all joins
- 15+ analysis queries covering store-wise sales, brand rankings, category profitability, staff performance, delayed shipments, inventory turnover, customer demographics, discount impact, and model year trends
- 5 reusable SQL Views: vw_sales_master, vw_staff_performance, vw_monthly_revenue, vw_inventory_status, vw_customer_ltv
- Single consolidated KPI summary query returning all headline metrics in one result set

Phase 3 — Python: Advanced Analytics & Segmentation

Python was used for exploratory data analysis, visualisation, and customer segmentation:

- Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn (KMeans, StandardScaler, silhouette_score)
- 18-section Jupyter Notebook covering all business use cases with supporting charts
- KMeans clustering applied with Elbow Method and Silhouette Score analysis to determine optimal k=4 clusters
- Customer segments identified: Champions, High Value, Mid Value, Low Value — each with distinct revenue and order frequency profiles
- Output: 5 insight CSV files exported for further use in Power BI or SQL

4. Analysis Results

4.1 Store-wise Revenue Performance

The three stores demonstrated significantly different performance levels across the analysis period:

Store Name	City / State	Completed Orders	Net Revenue	Revenue Share
Baldwin Bikes	New York	1,019	\$4,701,210	70.6%
Santa Cruz Bikes	California	284	\$1,255,492	18.8%
Rowlett Bikes	Texas	142	\$705,914	10.6%

Baldwin Bikes in New York generates 70.6% of all company revenue with 1,019 completed orders. Santa Cruz Bikes and Rowlett Bikes together contribute the remaining 29.4%, indicating an opportunity to scale up the underperforming stores through targeted investment and staff development.

4.2 Category Profitability Analysis

Product categories were ranked by net revenue after discount deductions:

Category	Units Sold	Net Revenue	Revenue Share	Avg Discount
Mountain Bikes	1,607	\$2,486,420	37.3%	~10.6%
Road Bikes	449	\$1,327,156	19.9%	~10.6%
Cruisers Bicycles	1,865	\$866,524	13.0%	~10.6%
Electric Bikes	117	\$714,437	10.7%	~10.6%
Cyclocross Bicycles	214	\$621,843	9.3%	~10.6%
Children Bicycles	1,207	\$462,248	6.9%	~10.6%
Comfort Bicycles	859	\$183,988	2.8%	~10.6%

Mountain Bikes lead in net revenue (\$2.49M) while Cruisers Bicycles lead in units sold (1,865 units) at a significantly lower price point. This indicates a clear premium versus volume split in the product portfolio. Comfort Bicycles represent the smallest revenue contribution and may warrant a pricing or marketing review.

4.3 Brand Performance

Trek dominates the brand landscape with nearly 59% revenue share:

Brand	Net Revenue	Revenue Share	Rank
Trek	\$3,919,363	58.8%	1
Electra	\$575,721	8.6%	2
Surly	\$539,861	8.1%	3
Sun Bicycle	\$460,087	6.9%	4
Pure Cycles	\$375,974	5.6%	5

The top 5 brands account for approximately 89% of total net revenue. While Trek's dominance reflects strong consumer preference, it creates supply chain risk. Expanding promotion of mid-tier brands such as Haro and Strider could improve portfolio resilience.

4.4 Staff Performance

Staff performance was measured by completed orders handled and net revenue generated:

Staff Name	Store	Orders Handled	Net Revenue	Rank
Marcelene Boyer	Baldwin Bikes	399	\$2,405,218	1
Venita Daniel	Baldwin Bikes	333	\$1,560,491	2
Genna Serrano	Baldwin Bikes	287	\$735,501	3
Mireille Phung	Santa Cruz Bikes	146	\$643,756	4
Kali Vargas	Santa Cruz Bikes	138	\$611,736	5
Layla Terrell	Rowlett Bikes	142	\$705,914	6

Marcelene Boyer at Baldwin Bikes leads all staff with \$2.4M in revenue generated across 399 orders. Three of the top six performers are from Baldwin Bikes, correlating with that store's overall dominance. Rowlett Bikes staff (Layla Terrell) handles 142 orders but shows strong per-order revenue, suggesting high-value transactions.

4.5 Order Fulfilment & Delayed Shipments

- Total orders placed: 1,615
- Completed orders: 1,445 (89.5%)
- Rejected orders: 116 (7.2%)
- Pending/Processing: 54 (3.3%)
- Orders shipped after required date: 458 (28.3% of shipped orders)
- Maximum recorded delay: 13 days
- Average delay: 4.2 days

The high rate of delayed shipments (28.3%) is a significant operational concern. The majority of delays cluster in Baldwin Bikes, given its volume. A logistics review — including supplier lead times, warehouse processing times, and carrier SLAs — is recommended as a priority action.

4.6 Inventory Health

Stock Status	SKU Count	% of Inventory	Action Required
Out of Stock	25	2.7%	Immediate reorder required
Low Stock	141	15.0%	Reorder within 2 weeks
Moderate	361	38.4%	Monitor closely
Well Stocked	412	43.9%	No action needed

Of the 939 SKUs tracked across all stores, 25 are completely out of stock and 141 are at critically low levels (below 5 units). Combined, this represents 17.7% of inventory at risk of causing lost sales. An automated reorder system triggered at 5 units would significantly reduce stockout risk.

4.7 Monthly Revenue Trends

Revenue exhibited strong seasonality over the 2016–2018 period:

- Peak month: January 2018 — \$381,430
- Strong Q1 performance observed consistently across all three years
- Revenue grew year-on-year from 2016 to 2018 with occasional mid-year dips
- Holiday season (October–December) showed above-average order frequency

The upward trend in revenue across the three-year period confirms business growth. Seasonal planning — particularly pre-purchasing high-demand inventory before Q1 peaks — would further maximise revenue.

4.8 Customer Segmentation (KMeans Clustering)

Using KMeans clustering with k=4 (selected via Elbow and Silhouette Score analysis), customers were segmented into four distinct groups:

Segment	Customer Count	Avg Lifetime Value	Recommended Action
Champions	~85	High (\$3,000+)	VIP loyalty programme, early access to new products
High Value	~210	Mid-High (\$1,500–3,000)	Upsell premium categories, personalised offers
Mid Value	~680	Mid (\$500–1,500)	Cross-sell, email nurture campaigns
Low Value	~470	Low (< \$500)	Re-engagement discounts, entry-level product promotion

Champions and High Value customers represent approximately 20% of the customer base but drive the majority of revenue. Retaining these segments through personalised engagement is the highest-priority CRM action.

5. Key Insights & Recommendations

#	Insight Area	Finding & Recommendation
1	Revenue Concentration	Baldwin Bikes generates 70.6% of total revenue — geographic diversification is a strategic priority.
2	Brand Dependency	Trek accounts for 58.8% of net revenue. Supplier concentration risk should be mitigated.
3	Discount Management	Average 10.6% discount across all orders. High-discount categories erode margin — targeted review recommended.
4	Delayed Shipments	458 orders (28.3%) shipped after the required date. Logistics process improvement needed, especially in Baldwin Bikes.
5	Inventory Gaps	25 SKUs are out of stock and 141 at low stock levels — immediate automated reorder triggers required.
6	Seasonal Peaks	January 2018 was the peak revenue month (\$381K). Marketing budgets should align with seasonal demand.
7	Customer Lifetime Value	Top customers are repeat buyers from NY and CA — loyalty programs can significantly improve retention.
8	Category Opportunity	Mountain Bikes lead revenue but Cruiser Bicycles lead units sold — pricing strategy can be optimised.

6. Project Deliverables

#	Deliverable	Description	Status
1	Cleaned CSV Files (x9 + Master)	All 9 tables cleaned and preprocessed; 1 merged master file (4,722 x 37)	Completed
2	SQL Script	Full schema, data loading, 15+ queries, 5 Views, KPI summary query	Completed
3	Python Jupyter Notebook	18-section EDA, KPI analysis, trend charts, KMeans segmentation	Completed
4	Excel Workbook	11-sheet workbook with KPI dashboard, pivot tables, charts, conditional formatting	Completed
5	Final Report (this document)	Comprehensive PDF/DOCX documentation of all phases and findings	Completed
6	Power BI Dashboard (.pbix)	Interactive dashboard with KPIs, slicers, maps, and drill-down	Completed

7. Conclusion

This project successfully delivered a comprehensive, multi-tool retail analytics solution covering all phases specified in the project brief. The cleaned and structured data, combined with rigorous SQL querying, Python-based analysis, and Excel-based reporting, provides the retail company with actionable intelligence across all operational dimensions.

The primary findings can be summarised as follows:

- Revenue is highly concentrated in Baldwin Bikes (70.6%) and in Trek as a brand (58.8%) — diversification strategies should be explored
- Mountain Bikes are the highest-revenue category, warranting continued investment in product depth and inventory
- 28.3% of shipments were delayed beyond the required delivery date — logistics optimisation is the most urgent operational priority
- 17.7% of inventory SKUs are at risk of stockout — automated reorder triggers are needed immediately
- Top customer segments (Champions + High Value) drive disproportionate revenue — targeted retention programmes would deliver strong ROI
- January consistently peaks in revenue — seasonal inventory planning and marketing campaigns should be aligned accordingly

With the foundation established in this project, the company is positioned to implement predictive analytics in the next phase — including demand forecasting for high-turnover SKUs and churn prediction for mid-value customers.

End of Report